Chemistry, The Central Science, 11th edition
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and Bruce E. Bursten

Chapter 7
Periodic Properties of the Elements

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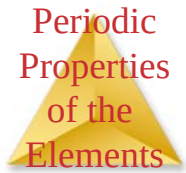

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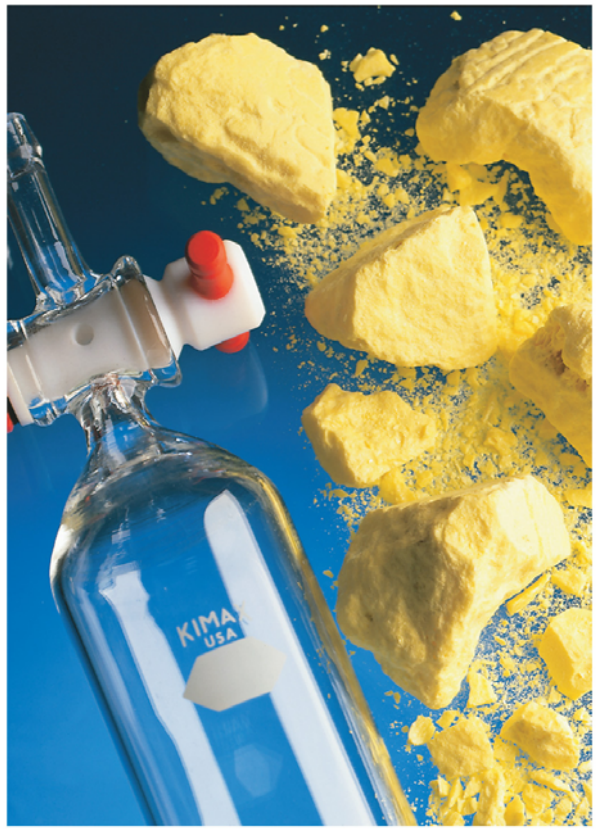
Development of Periodic Table

- Elements in the same group generally have similar chemical properties.
- Physical properties are not necessarily similar, however.



Periodic Properties of the Elements

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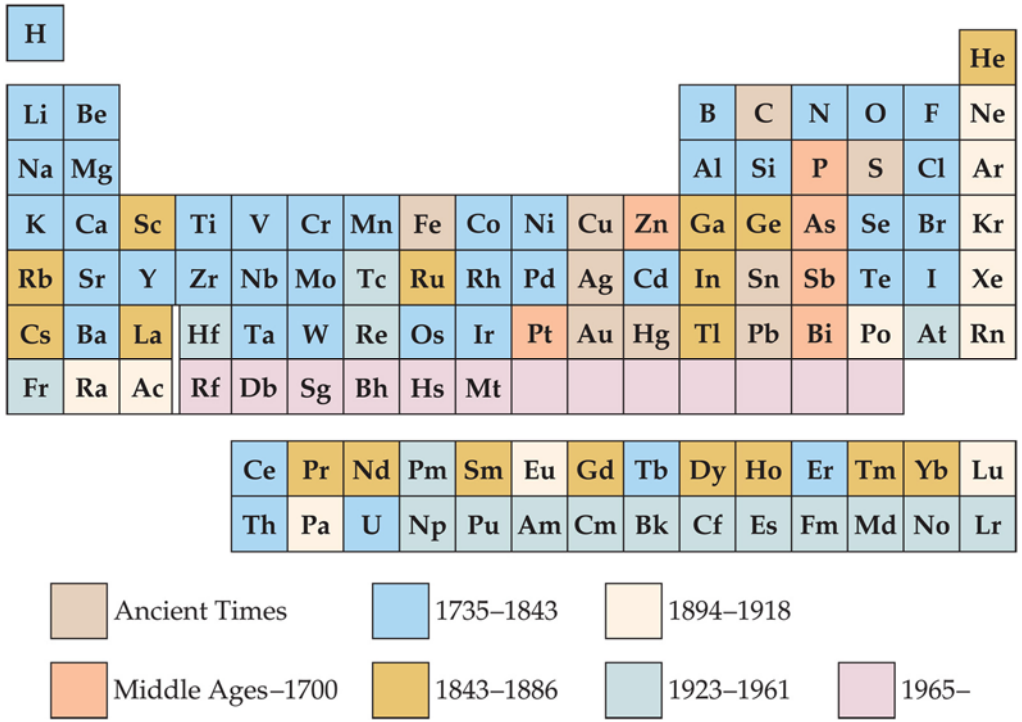
Development of Periodic Table



Dmitri Mendeleev and Lothar Meyer independently came to the same conclusion about how elements should be grouped.

Periodic Properties of the Elements

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Development of Periodic Table

Property	Mendeleev's Predictions for Eka-Silicon (made in 1871)	Observed Properties of Germanium (discovered in 1886)
Atomic weight	72	72.59
Density (g/cm^3)	5.5	5.35
Specific heat (J/g-K)	0.305	0.309
Melting point (°C)	High	947
Color	Dark gray	Grayish white
Formula of oxide	XO_2	GeO_2
Density of oxide (g/cm^3)	4.7	4.70
Formula of chloride	XCl_4	$GeCl_4$
Boiling point of chloride (°C)	A little under 100	84

Mendeleev, for instance, predicted the discovery of germanium (which he called eka-silicon) as an element with an atomic weight between that of zinc and arsenic, but with chemical properties similar to those of silicon.

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Periodic Trends

- In this chapter, we will rationalize observed trends in
  - Sizes of atoms and ions.
  - Ionization energy.
  - Electron affinity.

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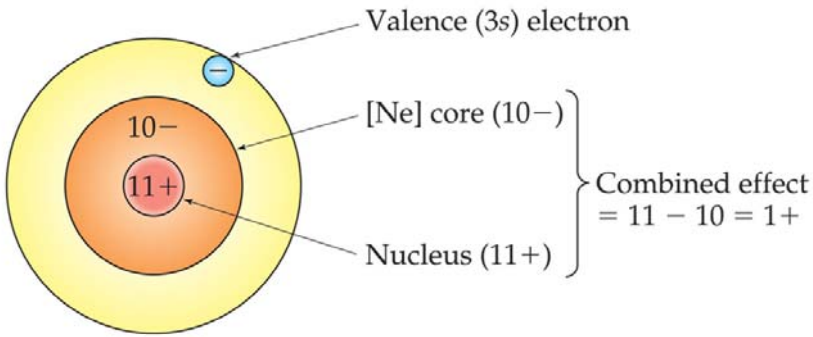
Effective Nuclear Charge



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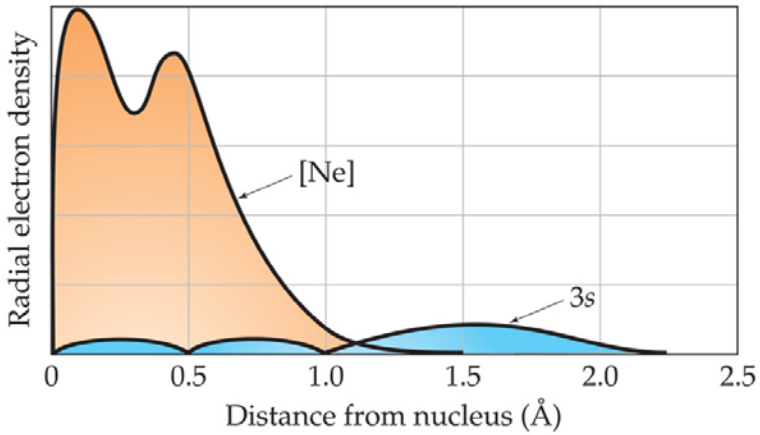

- In a many-electron atom, electrons are both attracted to the nucleus and repelled by other electrons.
- The nuclear charge that an electron experiences depends on both factors.

Periodic Properties of the Elements

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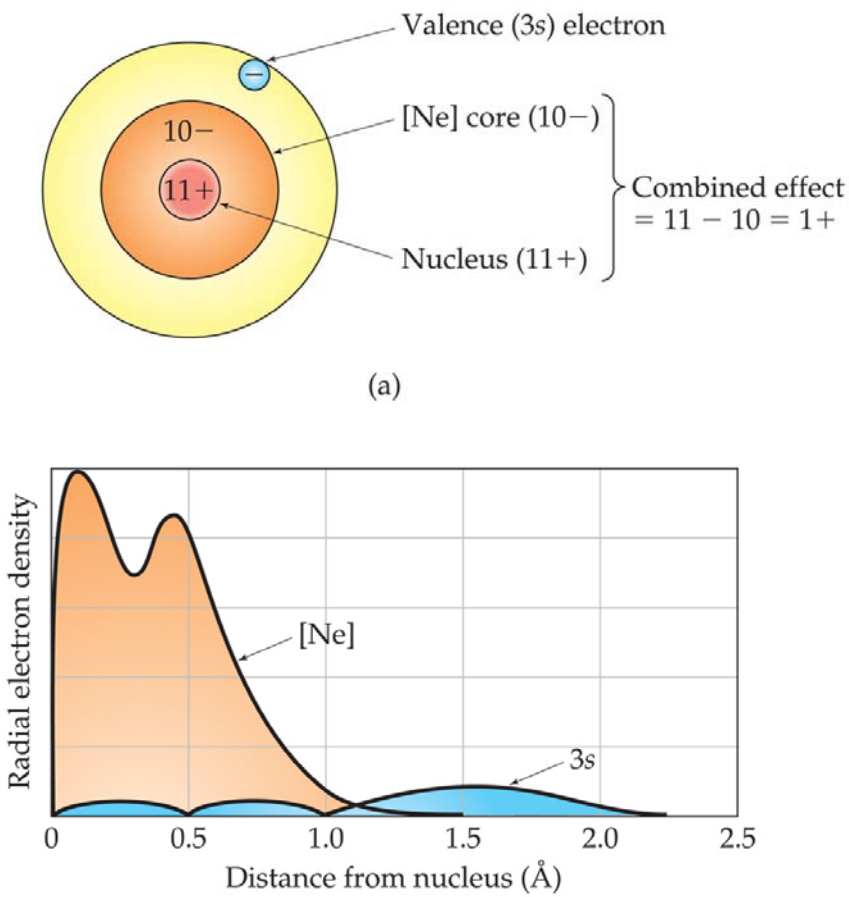
Effective Nuclear Charge



The effective nuclear charge, Z_{eff}, is found this way:

$$Z_{eff} = Z - S$$

where Z is the atomic number and S is a screening constant, usually close to the number of inner electrons.

Periodic Properties of the Elements

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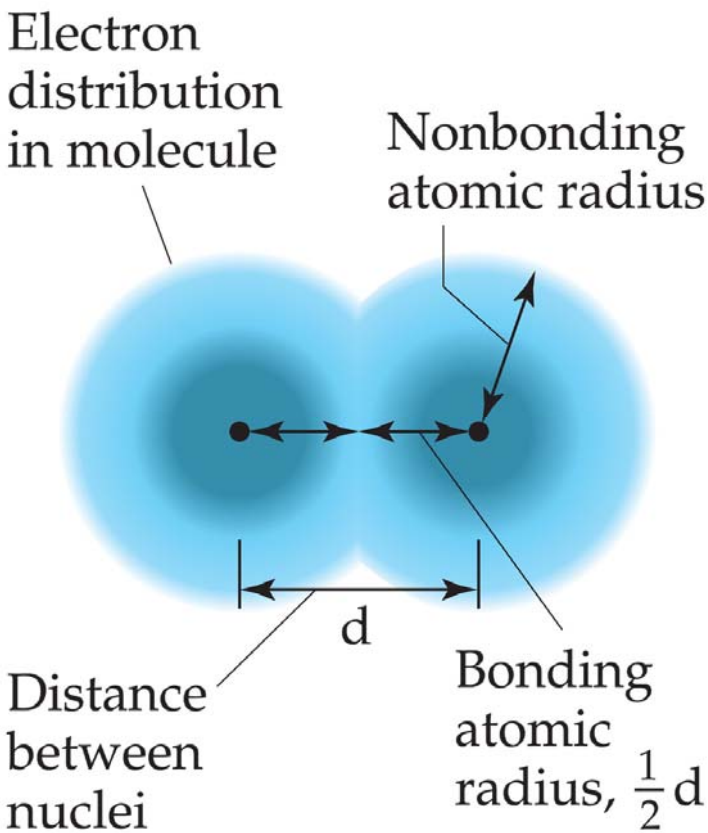
What Is the Size of an Atom?

The bonding atomic radius is defined as one-half of the distance between covalently bonded nuclei.



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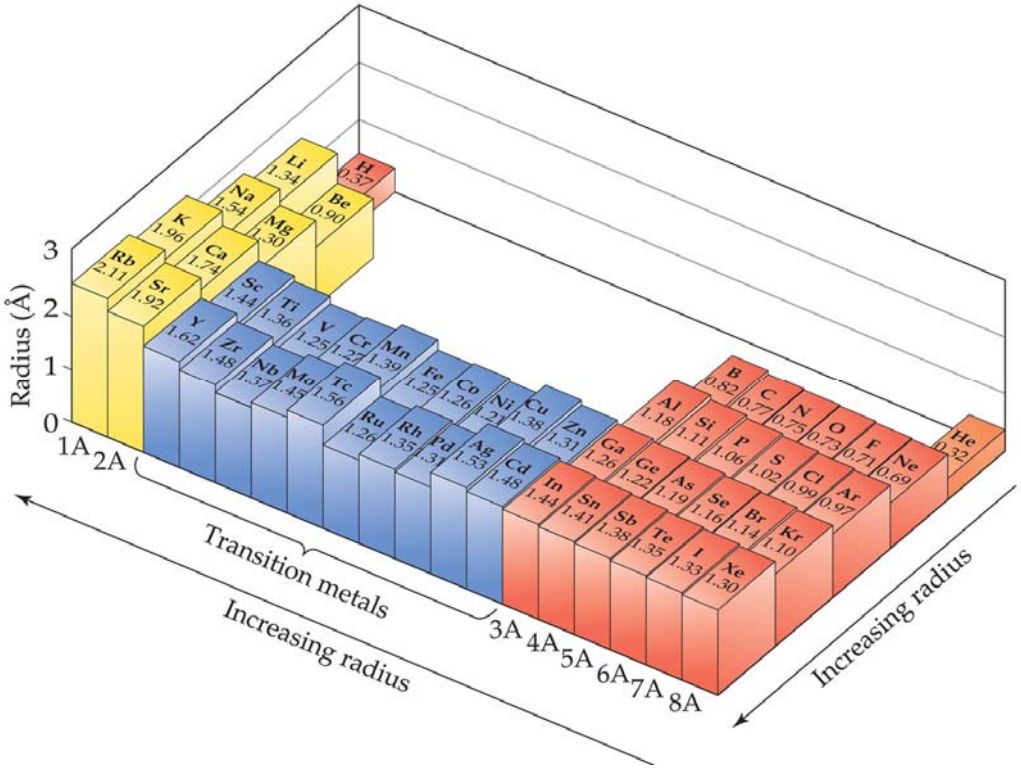
Sizes of Atoms



Bonding atomic radius tends to…

- …decrease from left to right across a row (due to increasing Z_{eff}).
- …increase from top to bottom of a column (due to increasing value of *n*).

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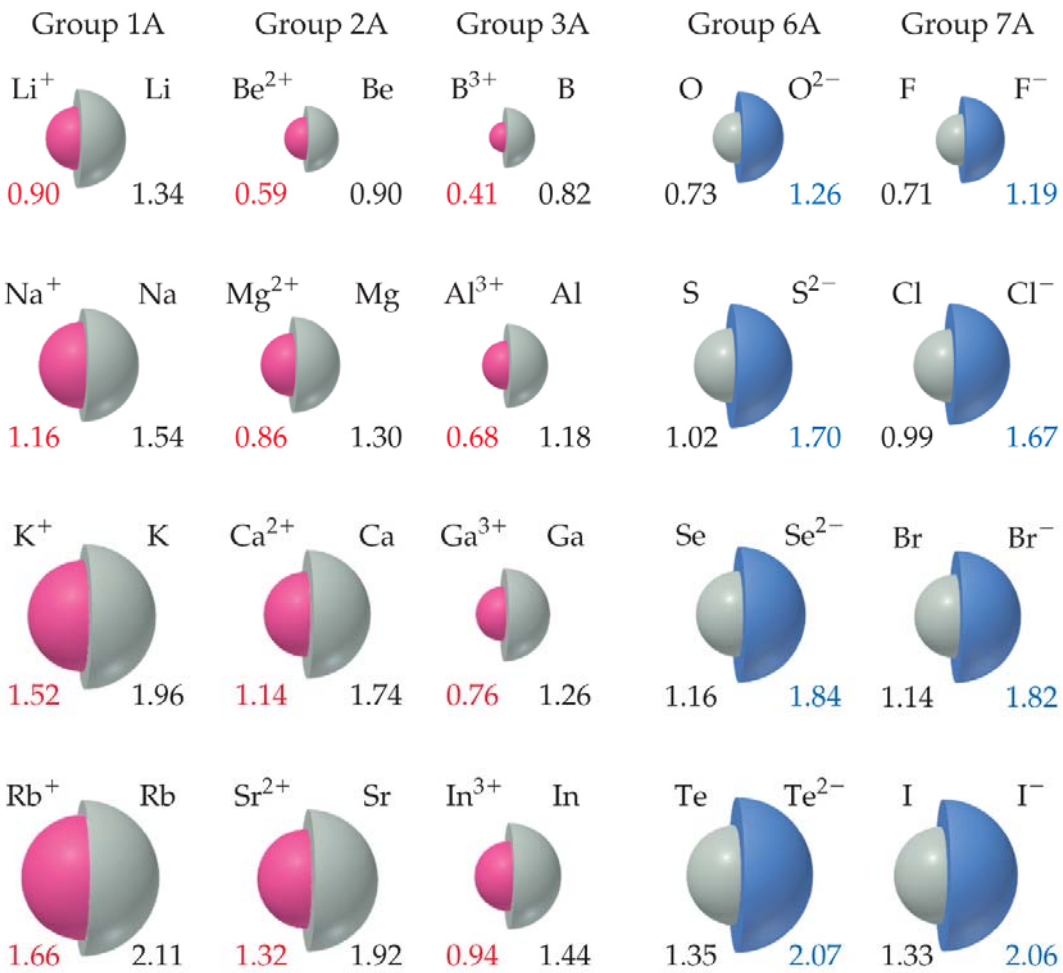
Sizes of Ions



- Ionic size depends upon:
  - The nuclear charge.
  - The number of electrons.
  - The orbitals in which electrons reside.

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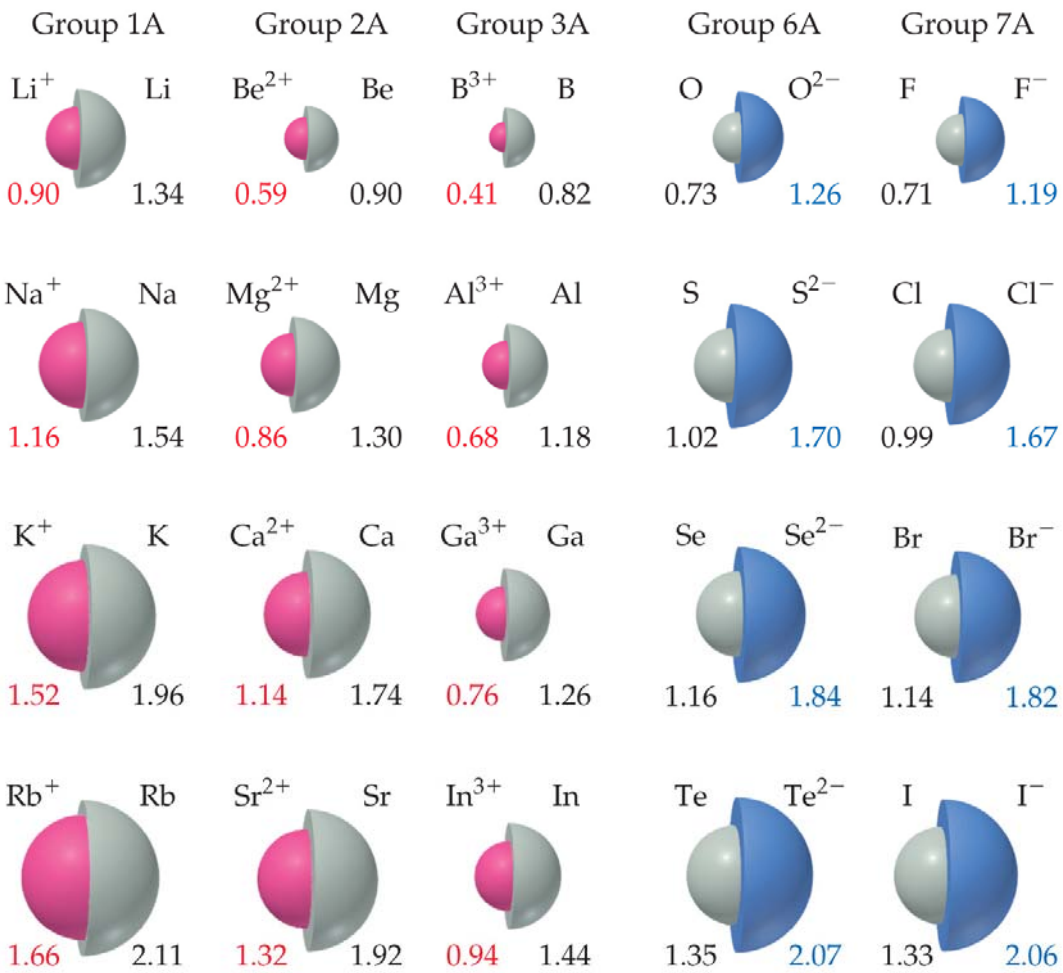
Sizes of Ions



- Cations are smaller than their parent atoms.
  - The outermost electron is removed and repulsions between electrons are reduced.

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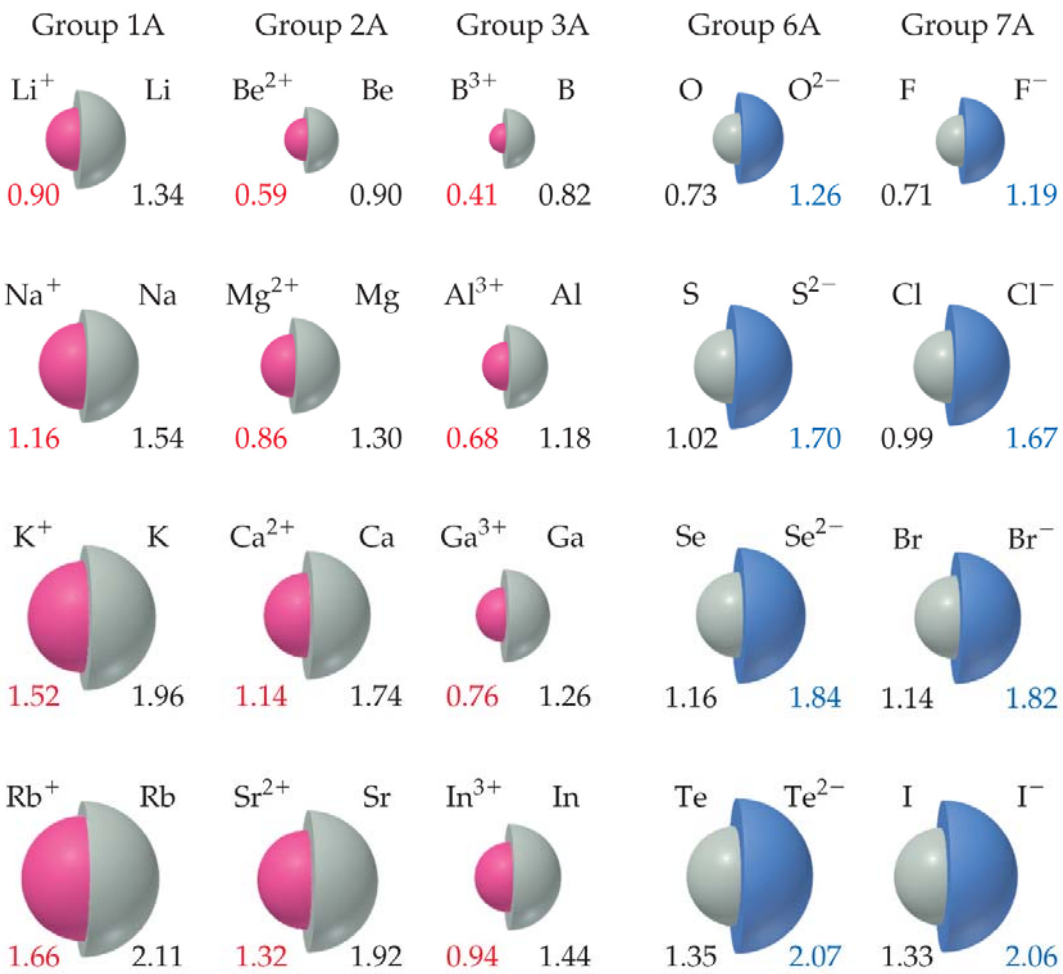
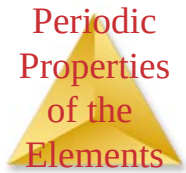
Sizes of Ions



- Anions are larger than their parent atoms.
  - Electrons are added and repulsions between electrons are increased.



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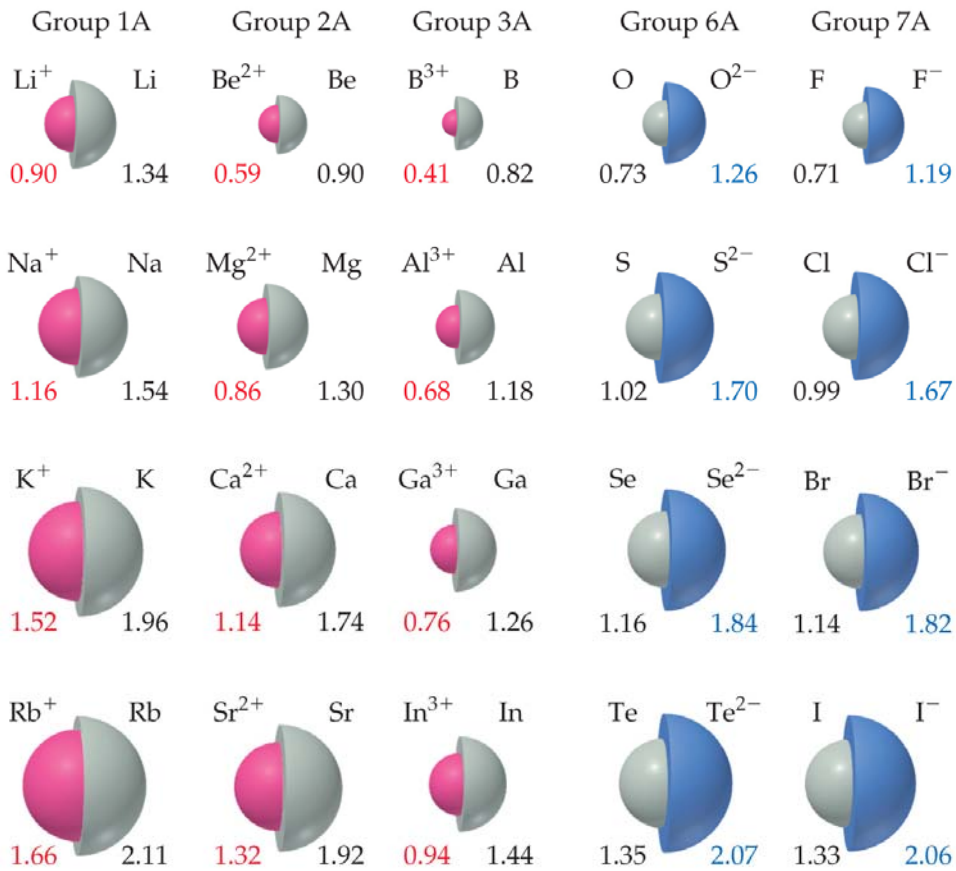
Sizes of Ions

- Ions increase in size as you go down a column.
  - This is due to increasing value of *n*.



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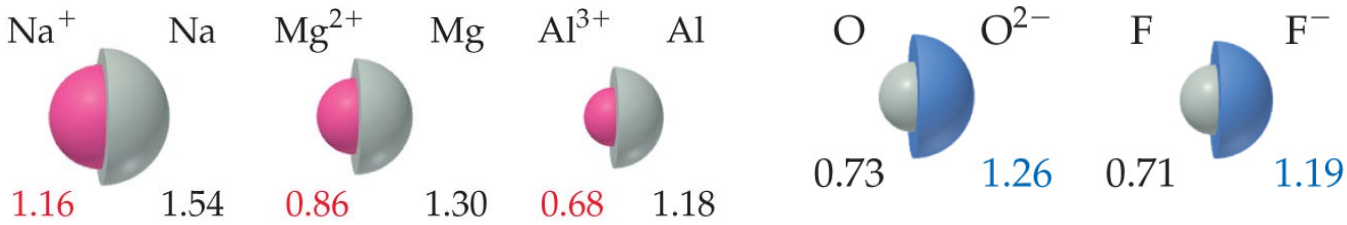
Sizes of Ions



- In an isoelectronic series, ions have the same number of electrons.
- Ionic size decreases with an increasing nuclear charge.

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Ionization Energy

- The ionization energy is the amount of energy required to remove an electron from the ground state of a gaseous atom or ion.
  - The first ionization energy is that energy required to remove first electron.
  - The second ionization energy is that energy required to remove second electron, etc.

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Ionization Energy

- It requires more energy to remove each successive electron.
- When all valence electrons have been removed, the ionization energy takes a quantum leap.

Element	I_1	I_2	I_3	I_4	I_5	I_6	I_7
Na	495	4562	(inner-shell electrons)				
Mg	738	1451	7733				
Al	578	1817	2745	11,577			
Si	786	1577	3232	4356	16,091		
P	1012	1907	2914	4964	6274	21,267	
S	1000	2252	3357	4556	7004	8496	27,107
Cl	1251	2298	3822	5159	6542	9362	11,018
Ar	1521	2666	3931	5771	7238	8781	11,995

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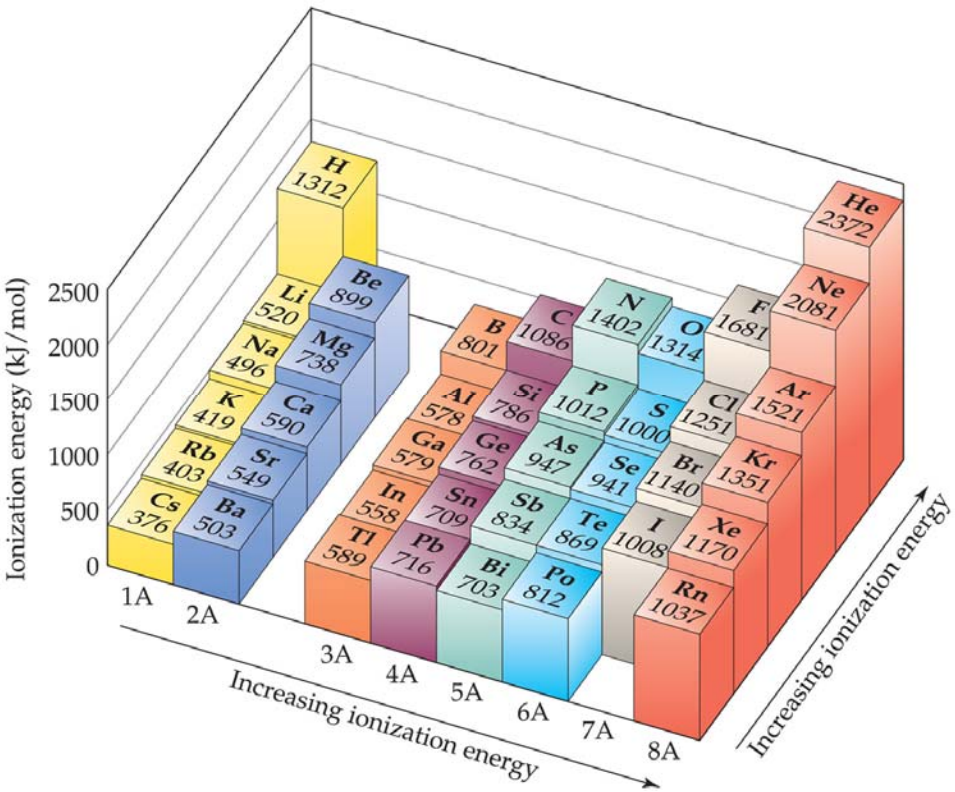
Trends in First Ionization Energies



- As one goes down a column, less energy is required to remove the first electron.
  - For atoms in the same group, Z_{eff} is essentially the same, but the valence electrons are farther from the nucleus.

Periodic Properties of the Elements

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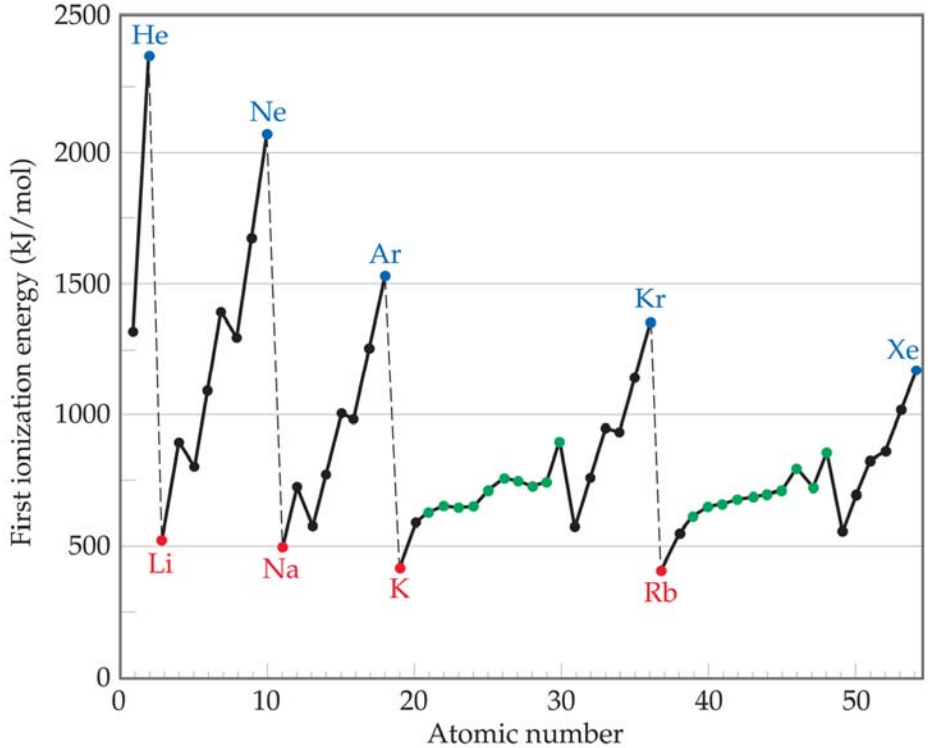
Trends in First Ionization Energies

- Generally, as one goes across a row, it gets harder to remove an electron.
  - As you go from left to right, Z_{eff} increases.



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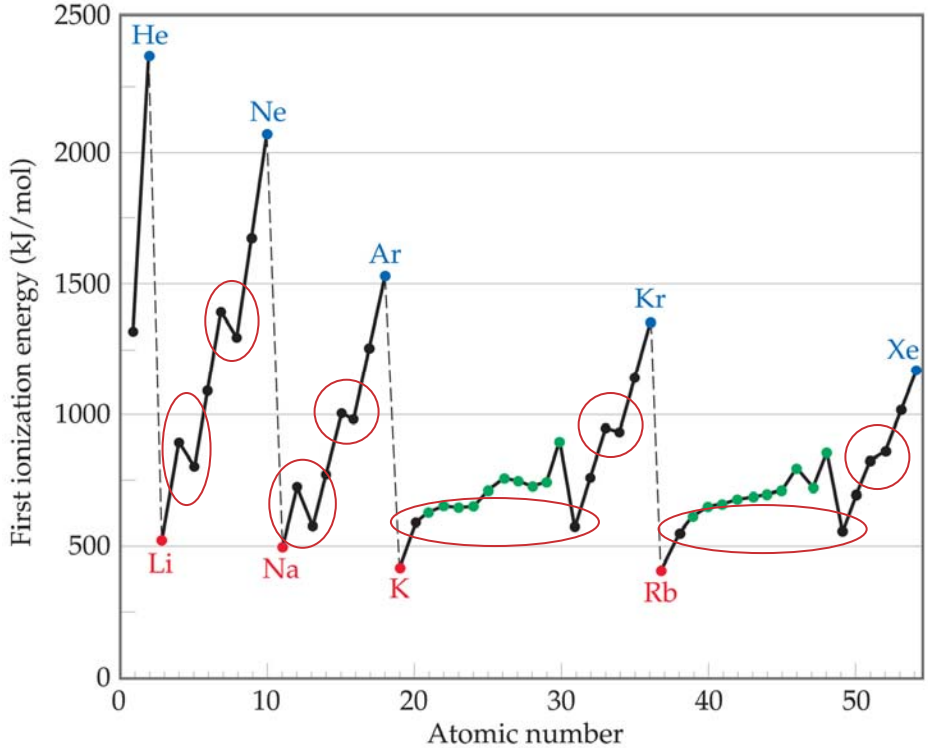
Trends in First Ionization Energies

However, there are two apparent discontinuities in this trend.



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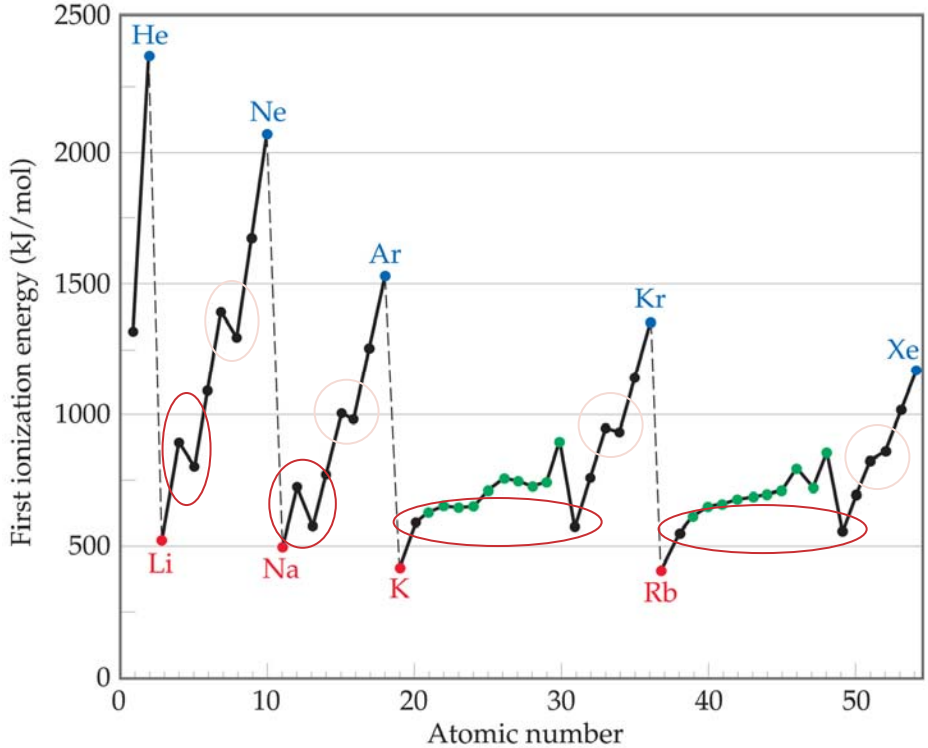
Trends in First Ionization Energies

- The first occurs between Groups IIA and IIIA.
- In this case the electron is removed from a *p*-orbital rather than an *s*-orbital.
  - The electron removed is farther from nucleus.
  - There is also a small amount of repulsion by the *s* electrons.



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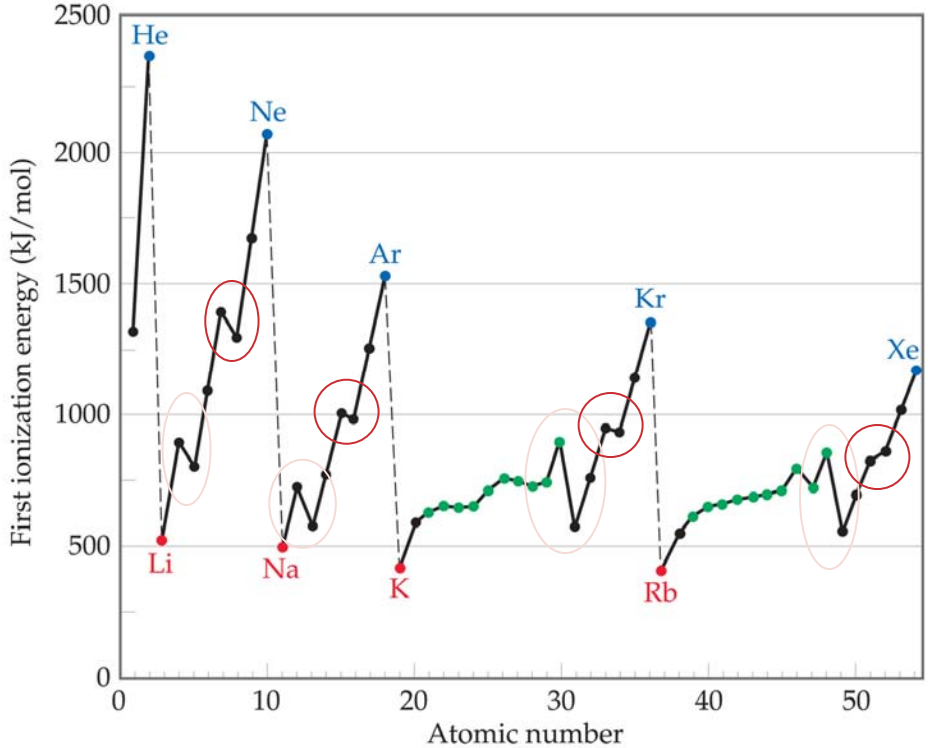
Trends in First Ionization Energies

- The second occurs between Groups VA and VIA.
  - The electron removed comes from doubly occupied orbital.
  - Repulsion from the other electron in the orbital aids in its removal.



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Electron Affinity

Electron affinity is the energy change accompanying the addition of an electron to a gaseous atom:

$$Cl + e^- \longrightarrow Cl^-$$

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Trends in Electron Affinity

H −73							He > 0
Li −60	Be > 0	B −27	C −122	N > 0	O −141	F −328	Ne > 0
Na −53	Mg > 0	Al −43	Si −134	P −72	S −200	Cl −349	Ar > 0
K −48	Ca −2	Ga −30	Ge −119	As −78	Se −195	Br −325	Kr > 0
Rb −47	Sr −5	In −30	Sn −107	Sb −103	Te −190	I −295	Xe > 0
1A	2A	3A	4A	5A	6A	7A	8A

In general, electron affinity becomes more exothermic as you go from left to right across a row.

Periodic Properties of the Elements

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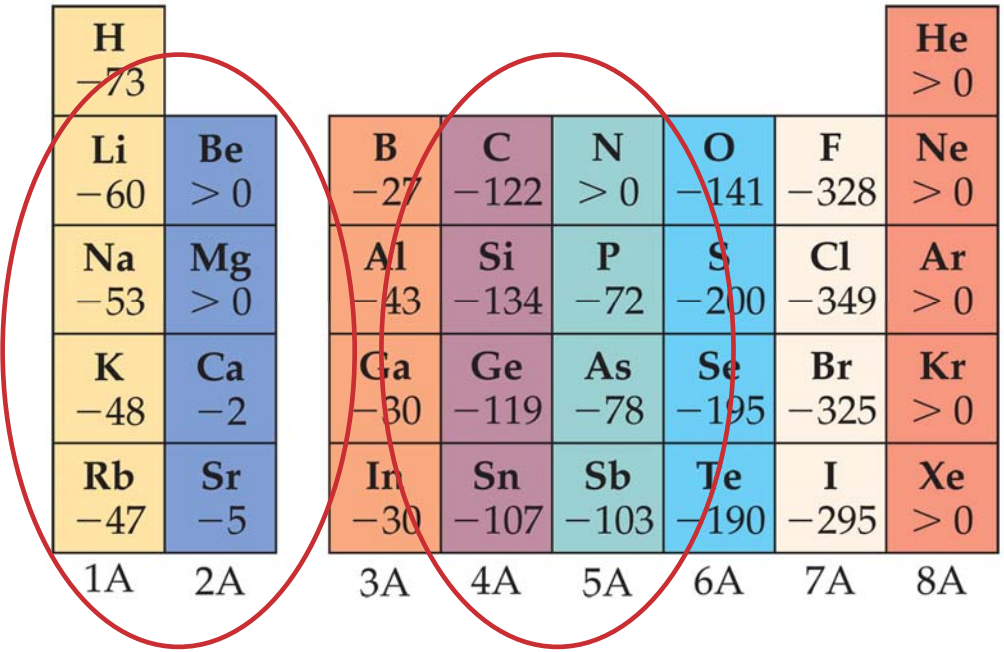
Trends in Electron Affinity



There are again, however, two discontinuities in this trend.

Periodic Properties of the Elements

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Trends in Electron Affinity

H −73							He > 0
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Rb −47	Sr −5	In −30	Sn −107	Sb −103	Te −190	I −295	Xe > 0
1A	2A	3A	4A	5A	6A	7A	8A

- The first occurs between Groups IA and IIA.
  - The added electron must go in a *p*-orbital, not an *s*-orbital.
  - The electron is farther from nucleus and feels repulsion from the *s*-electrons.

Periodic Properties of the Elements

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Trends in Electron Affinity

H −73							He > 0
Li −60	Be > 0	B −27	C −122	N > 0	O −141	F −328	Ne > 0
Na −53	Mg > 0	Al −43	Si −134	P −72	S −200	Cl −349	Ar > 0
K −48	Ca −2	Ga −30	Ge −119	As −78	Se −195	Br −325	Kr > 0
Rb −47	Sr −5	In −30	Sn −107	Sb −103	Te −190	I −295	Xe > 0
1A	2A	3A	4A	5A	6A	7A	8A

- The second occurs between Groups IVA and VA.
  - Group VA has no empty orbitals.
  - The extra electron must go into an already occupied orbital, creating repulsion.

Periodic Properties of the Elements

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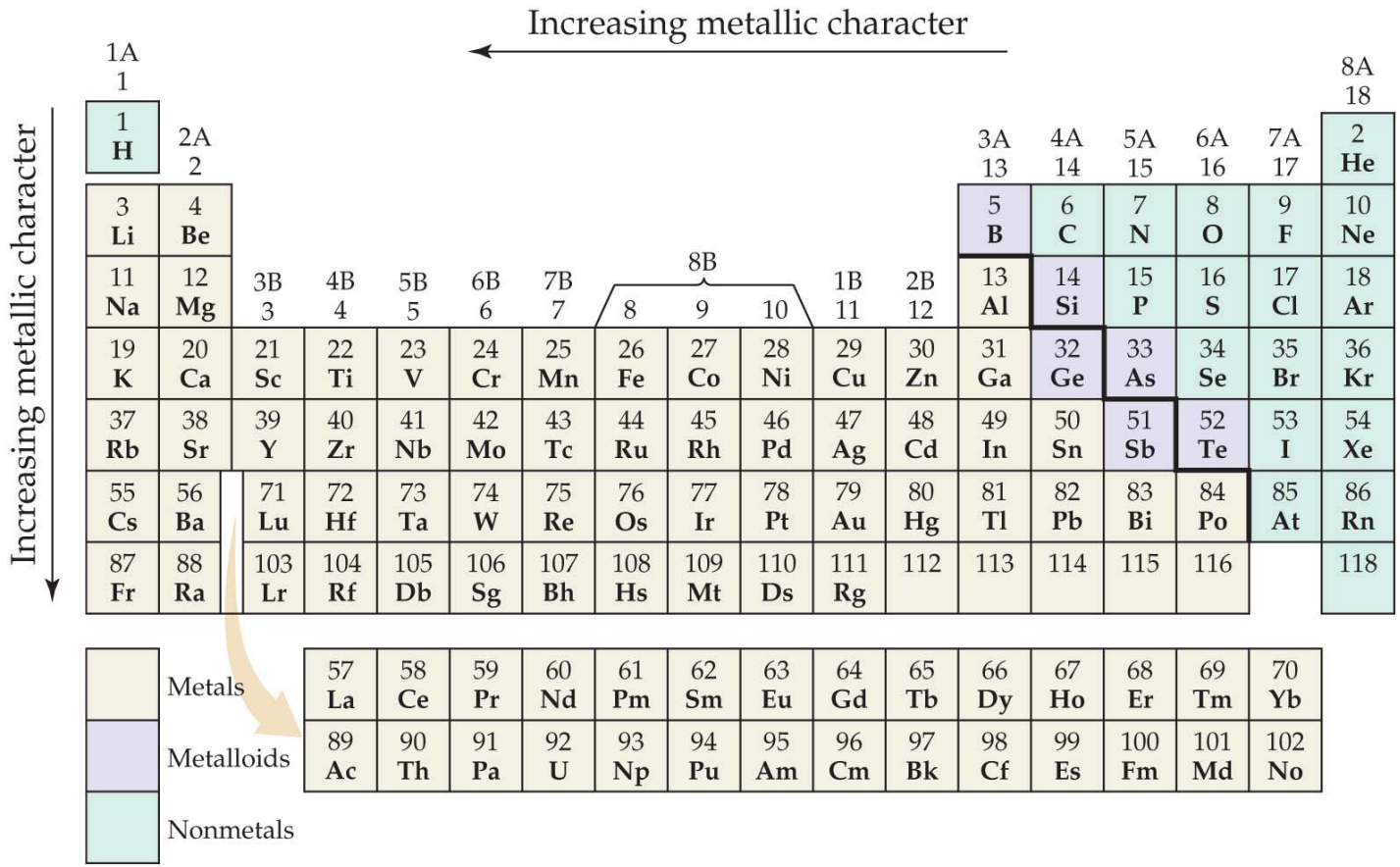
Properties of Metal, Nonmetals, and Metalloids



Periodic Properties of the Elements

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Metals versus Nonmetals

Metals	Nonmetals
Have a shiny luster; various colors, although most are silvery	Do not have a luster; various colors
Solids are malleable and ductile	Solids are usually brittle; some are hard, some are soft
Good conductors of heat and electricity	Poor conductors of heat and electricity
Most metal oxides are ionic solids that are basic	Most nonmetal oxides are molecular substances that form acidic solutions
Tend to form cations in aqueous solution	Tend to form anions or oxyanions in aqueous solution

Differences between metals and nonmetals tend to revolve around these properties.

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Metals versus Nonmetals

- Metals tend to form cations.
- Nonmetals tend to form anions.

1A	2A	Transition metals										3A	4A	5A	6A	7A	8A
H^+																H^-	NOBLE GASES
Li^+														N^{3-}	O^{2-}	F^-	
Na^+	Mg^{2+}											Al^{3+}		P^{3-}	S^{2-}	Cl^-	
K^+	Ca^{2+}	Sc^{3+}	Ti^{4+}	V^{5+} V^{4+}	Cr^{3+}	Mn^{2+} Mn^{4+}	Fe^{2+} Fe^{3+}	Co^{2+} Co^{3+}	Ni^{2+}	Cu^+ Cu^{2+}	Zn^{2+}				Se^{2-}	Br^-	
Rb^+	Sr^{2+}								Pd^{2+}	Ag^+	Cd^{2+}		Sn^{2+} Sn^{4+}	Sb^{3+} Sb^{5+}	Te^{2-}	I^-	
Cs^+	Ba^{2+}								Pt^{2+}	Au^+ Au^{3+}	Hg_2^{2+} Hg^{2+}		Pb^{2+} Pb^{4+}	Bi^{3+} Bi^{5+}			

Periodic Properties of the Elements

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Metals

They tend to be lustrous, malleable, ductile, and good conductors of heat and electricity.

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Metals

- Compounds formed between metals and nonmetals tend to be ionic.
- Metal oxides tend to be basic.

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Nonmetals

- These are dull, brittle substances that are poor conductors of heat and electricity.
- They tend to gain electrons in reactions with metals to acquire a noble gas configuration.

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Nonmetals

- Substances containing only nonmetals are molecular compounds.
- Most nonmetal oxides are acidic.

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Metalloids

- These have some characteristics of metals and some of nonmetals.
- For instance, silicon looks shiny, but is brittle and fairly poor conductor.

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